

CUSTOMER ANALYTICS · PORTFOLIO CASE STUDY

Customer Shopping Behavior Analysis

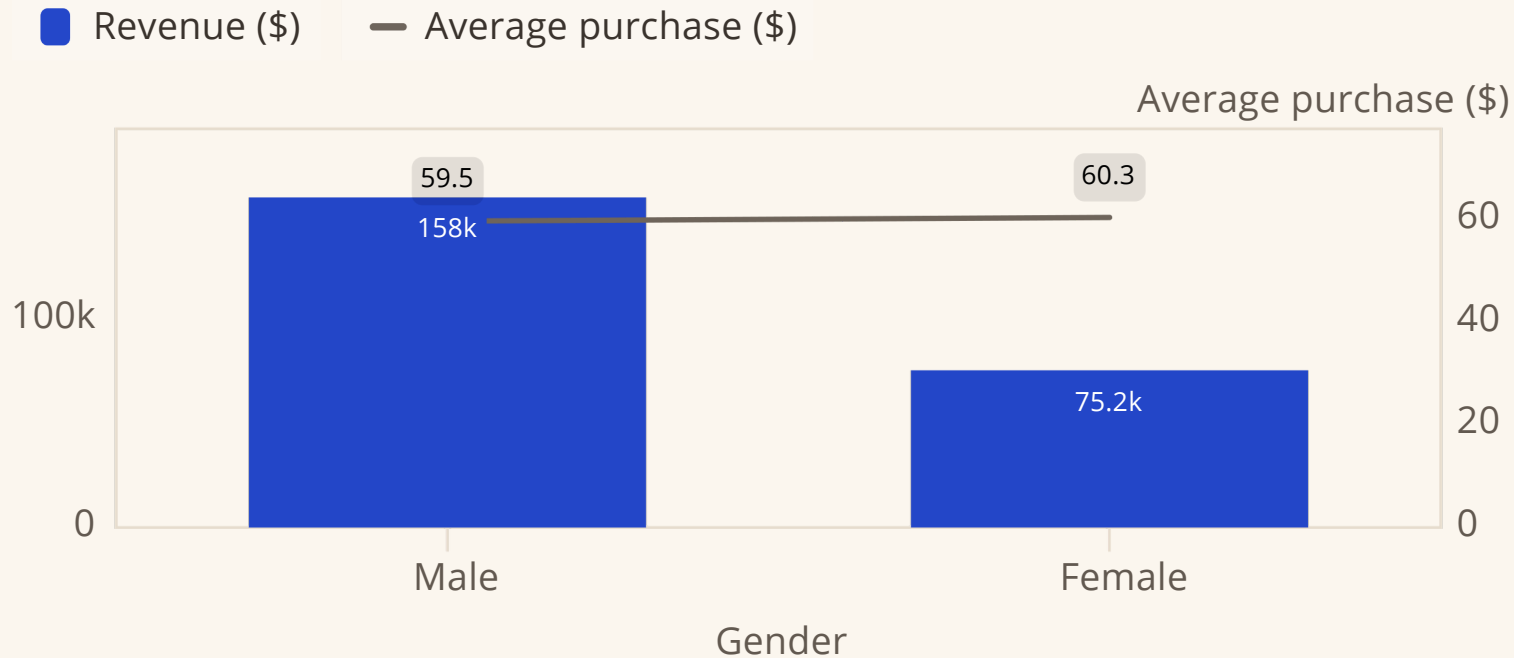
PYTHON · MYSQL · SQL · POWER BI

From 3,900 retail transactions to practical growth recommendations.



Revenue is shaped by volume, not basket size

Men generate more revenue because they represent more customers; average purchase is nearly flat.



\$157,890

MALE REVENUE
68% of customers.

\$75,191

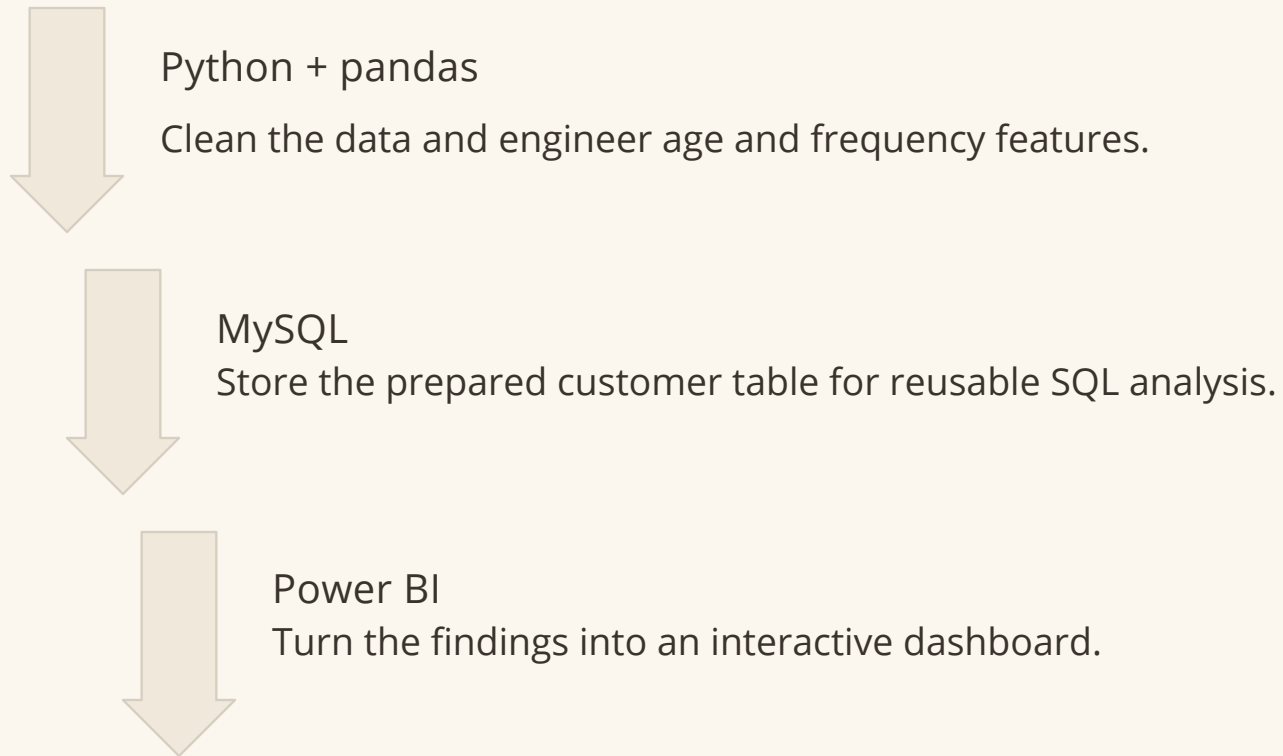
FEMALE REVENUE
32% of customers.

Average purchase

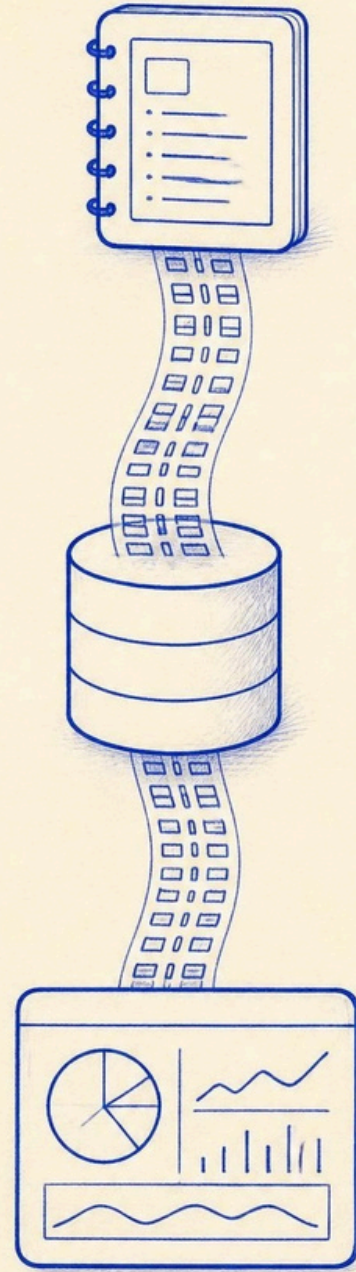
\$59.54 for men versus \$60.25 for women.

Revenue follows customer volume while average purchase stays nearly flat.

A complete analytics pipeline turns raw rows into decisions



Project outputs: a notebook, ten SQL business questions, and a single-page dashboard.



Cleaning made the dataset analysis-ready

Small structural decisions made the customer behavior easier to compare.

01 IMPUTE

37

Missing review ratings

Filled with category-level medians.

02 REMOVE

1

Redundant field

Promo Code Used was removed because it duplicated Discount Applied in every row.

03 ENGINEER

2

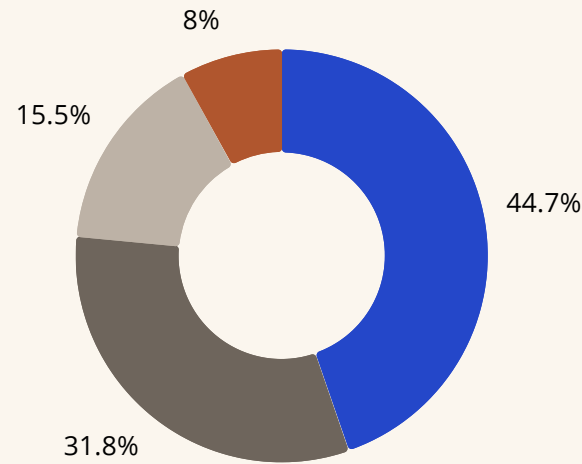
Behavior lenses

Age groups and purchase frequency in days made customer behavior easier to compare.

Clothing and Accessories carry the business

Two categories provide over three quarters of revenue, with clear products to feature.

Clothing Accessories Footwear Outerwear



\$104,264

CLOTHING · 44.7%
The largest revenue category.

31.8%

ACCESSORIES
Together, the two categories provide over three quarters of revenue.

Clothing and Accessories dominate revenue.

Feature: Pants, Blouse, Jewelry, Sunglasses, and Belt.

SUBSCRIPTION VALUE

Subscriptions are not yet changing customer value

The offer reaches too few customers and currently creates almost no difference in purchase value.

PURCHASE VALUE

\$59.49

Average purchase, versus \$59.87 for non-subscribers.

REACH

27%

Customers subscribe

The current subscription offer is not yet broad-based.

REPEAT BUYERS

27.6%

Subscribe

Nearly the same as the 27.0% overall rate.

SOURCE: SUBSCRIBER AND NON-SUBSCRIBER PURCHASE BEHAVIOR

PROMOTION EFFECTIVENESS

Discounts are widespread, but spend is slightly lower

Discounting is common, yet the average purchase is marginally lower when a discount is applied.

DISCOUNT COVERAGE

43%

of purchases use a discount

Discounting is already a broad part of the purchase journey.

AVERAGE PURCHASE

\$59.28

 discounted

\$60.13

 without discount

Spend is slightly lower with a discount.

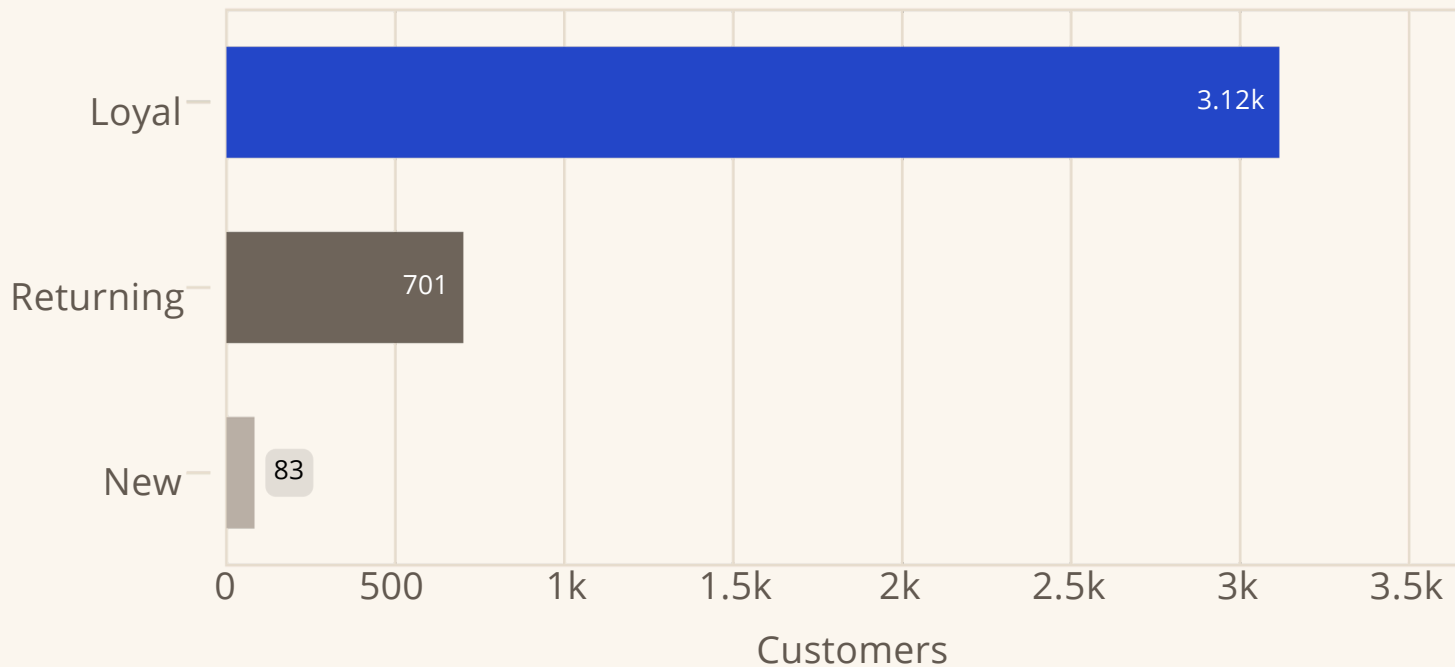
Frequently discounted: Hat, Sneakers, Coat, Sweater, and Pants — roughly half of sales.

SOURCE: DISCOUNT APPLIED AND PURCHASE AMOUNT ANALYSIS

Loyalty is strong, but the subscription funnel misses it

The customer base is highly loyal; the largest opportunity is converting repeat buyers who have not subscribed.

Segment



3,116

LOYAL CUSTOMERS

79.9% have 11 or more previous purchases.

701

RETURNING

A sizeable audience beyond the new-customer funnel.

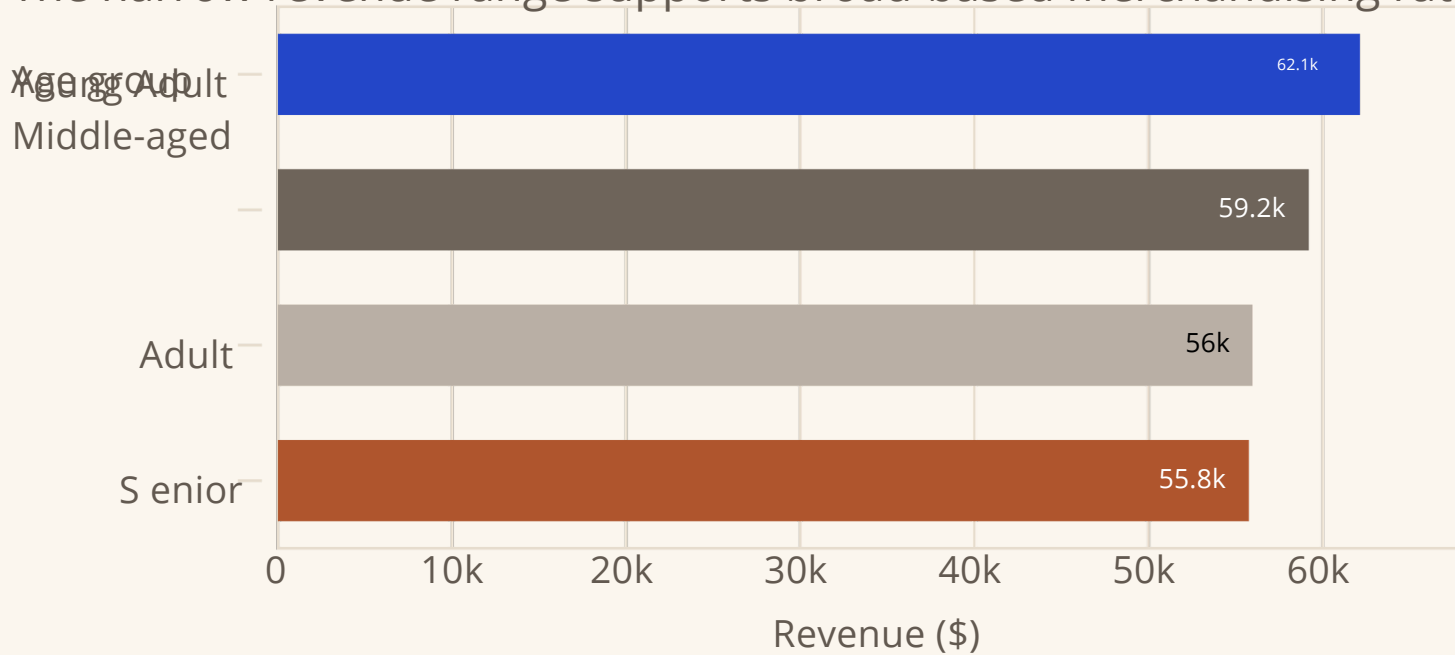
Opportunity: convert repeat buyers who have not subscribed.

Customer base by previous purchases.

SOURCE: CUSTOMER SEGMENTS BY PREVIOUS PURCHASES

Four age groups contribute almost equally

The narrow revenue range supports broad-based merchandising rather than one age-led strategy.



\$62,143

YOUNG ADULT

26.7% of revenue.

Middle-aged \$59,197

Adult \$55,978

Senior \$55,763

Broad-based merchandising beats a single age-led strategy.

Revenue is distributed evenly across age groups.

SOURCE: REVENUE BY AGE GROUP

Make loyalty more valuable and growth more targeted

The next moves focus on subscription value, disciplined promotion, customer acquisition, and category strength.

01 Redesign the subscription offer

Test benefits such as free Express shipping or early access.

02 Reduce discounts selectively

Test reducing discounts on products already discounted about half the time.

03 Target women with acquisition campaigns

Spend per purchase is comparable, but customer share is smaller.

04 Protect the category engine

Protect Clothing and Accessories, feature best sellers, and promote high-rated Gloves, Sandals, and Boots.